

Optimization Models and Methods: Group 62 Project 2 Final Report

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1 Introduction

Scheduling professional sports leagues is a classic and highly relevant operations research problem. In the context of the National Basketball Association (NBA), the schedule must satisfy a large number of operational requirements, including fixed game dates, home and away allocations, opponent frequencies, and fairness considerations across teams. Because the NBA regular season consists of a dense schedule spread across multiple time zones, even minor changes in sequencing can have meaningful impacts on player fatigue, travel burden, and competitive balance.

The objective of this project was to model and analyze the NBA scheduling problem using integer programming techniques. The investigation was divided into multiple stages. First, the existing schedule data was processed to recover the implied home dates, away dates, and matchup frequencies for each team. Second, these outputs were used to formulate a feasibility integer program that reconstructs a valid NBA schedule. Third, an additional operational constraint related to time-zone travel was introduced to study the feasibility under more realistic travel considerations. Finally, the schedule was improved through an optimization model that minimizes the maximum travel burden experienced by any single team.

What makes this problem challenging is that all the constraints are dependent upon each other. Making any changes to even a single constraint will result in changes to several others, which makes the problem difficult to solve manually and requires an optimization model.

This problem has strong real-world applications in sports analytics, league operations, and scheduling optimization. Sports leagues regularly use mathematical optimization to construct schedules that balance logistics, fairness, and broadcast requirements. Similar methodologies are widely used in airline crew scheduling, workforce planning, and tournament design. The project demonstrates how mathematical programming can be used not only to reproduce an existing schedule but also to diagnose infeasibility under stricter operating rules and propose improved schedules that reduce travel inequality.

2 Data Processing and Exploratory Analysis

Before formulating the optimization model, data preparation is done to identify the relevant scheduling data from the input dataset. In this dataset, the home team, the away team, and

the date of the match have been provided for all the matches.

Home and away game information was extracted from the data set through filtering depending on whether there is a team in either the Home or Visitor columns. The number of match-ups for each ordered pair was extracted by counting the frequency of occurrence of each pair.

The computation of the dates where the teams played at home and those where they played away are carried out for each of the teams. Moreover, the frequencies of each team playing against every other team are also computed for each team. Both home and away frequencies will be taken into account.

It is ensured that the optimization model maintains the properties of the schedule like the home-away balance and frequencies of playing against each other. This also helps in checking the consistency of the dataset itself, since the frequencies of home and away matches should match.

3 Feasibility Model Formulation

We formulated the problem as a binary integer program. The decision variable is:

$$x_{i,j,d} = \begin{cases} 1 & \text{if team } i \text{ plays at home against team } j \text{ on date } d \\ 0 & \text{otherwise} \end{cases}$$

This variable fully determines the schedule, as it specifies which teams play each other, where the game is played, and on which date.

3.1 Constraints

The model enforces that each team plays at home exactly on its designated home dates. This constraint ensures that each team plays at home exactly on the dates specified in the original schedule. It prevents the model from shifting home games to different dates:

$$\sum_{j \neq i} x_{i,j,d} = \text{home_on}_{i,d} \quad \forall i, d$$

Similarly, each team must play away exactly on its designated away dates. This constraint complements the home constraint by enforcing that each team plays away exactly on its des-

ignated away dates, ensuring that every team participates in exactly one game per scheduled date:

$$\sum_{j \neq i} x_{j,i,d} = \text{away_on}_{i,d} \quad \forall i, d$$

The total number of home games between each pair of teams must match the original schedule:

$$\sum_d x_{i,j,d} = \text{home_vs}_{i,j} \quad \forall i \neq j$$

Likewise, the number of away games must also match. These constraints preserve the competitive structure of the league by ensuring that each pair of teams plays the same number of games as in the original schedule:

$$\sum_d x_{j,i,d} = \text{away_vs}_{i,j} \quad \forall i \neq j$$

We also include a constraint to prevent teams from playing themselves. This constraint eliminates invalid matchups where a team would play against itself:

$$x_{i,i,d} = 0 \quad \forall i, d$$

3.2 Objective Function

Since the objective of this stage is to identify feasible schedules rather than optimize a specific metric, the objective function is defined as a constant. This means that the solver is solely focused on the constraints. The objective function will therefore be formulated as follows:

$$\min 0$$

The goal is simply to determine whether a schedule exists that satisfies all constraints.

4 Extended Model with Travel Constraints

A further way to account for real scheduling concerns is by adding a restriction that would reduce long travel between subsequent games. Every arena has an associated numeric time zone parameter. For every team, we consider sets of three successive games and calculate the time zone change during those sets.

Let $tz(i, d)$ be the time zone parameter at which team i has a game scheduled on day d . For any set of three successive games d_1, d_2, d_3 , it is required that

$$|tz(i, d_1) - tz(i, d_2)| + |tz(i, d_2) - tz(i, d_3)| < 4$$

This implies that any sequence which does not meet this requirement is automatically removed from the feasible domain using further constraints. This greatly narrows down the number of feasible schedules and adds another level of difficulty to the process. It guarantees that the teams do not undergo long distances between two successive matches. For instance, it ensures that there will be no schedule whereby teams travel from the East to the West and then return back to the East.

5 Schedule Improvement Model

In order to improve the scheduling algorithm even more, we include travel distance within our model. We use arena distance information to calculate the travel distance from one game to another for each individual team.

Let $dist(i, d)$ represent the travel distance for team i on date d . The objective is to minimize the maximum total travel distance across all teams:

$$\min \max_i \sum_d dist(i, d)$$

This formulation balances travel across teams and aims to reduce the overall travel burden, resulting in a more efficient and practical schedule.

6 Results and Discussion

This means that the feasibility model successfully recognizes schedules that respect all constraints that arise from the structure of the data set provided. Therefore, it seems to represent the nature of such scheduling problems very well by taking all relevant variables into consideration.

Adding the traveling constraint makes the scheduling problem significantly harder, as there are fewer possible solutions. Moreover, the complexity of solving this problem increases,

which emphasizes the sensitivity of scheduling problems when additional restrictions apply, especially in case of geographical aspects of the problem.

Furthermore, adding the traveling distance to the objective function of the optimization process provides valuable insight into how scheduling can improve overall performance through optimization, since schedules generated after adding this constraint reduce the traveling effort and distribute it across more teams than before.

In summary, the findings suggest that scheduling problems tend to be rather sensitive to changes made to the problem because of the multitude of different constraints involved.

7 Conclusion & Executive Summary

The project proposes integer programming models for solving the scheduling problem in the National Basketball Association (NBA). Using structural patterns derived from the datasets, we were able to build a feasibility model that reproduces feasible schedules by respecting both home/away game patterns and matchups.

The inclusion of travel constraints shows how sensitive scheduling problems are to the realities involved in the process. Adding a constraint that minimizes changes in time zones severely limits the number of feasible solutions. This implies that although there exist schedules which meet all the necessary structural conditions, very few remain once other factors such as distance are considered.

In extending the model to include travel distance, it becomes clear why optimizing models are crucial in solving scheduling problems. Going beyond feasibility offers one a way of improving on existing schedules in terms of efficiency.

All in all, the project illustrates that scheduling is a very complicated process where small changes in the constraints used result in huge variations in feasibility. Through integer programming, however, it is possible to efficiently solve this complex scheduling problem.

8 Bibliography